

AI acknowledgement coversheet

DECO7230 Digital Prototyping and Extended Reality

Interactive Prototype Outcome Evaluation 2

Name	Zixu Yang	Student Number	48939559
Course	DECO7230	Assignment	Interactive Prototype Outcome Evaluation 2

AI use

☒ **Yes, AI tools have been used to generate or edit material in this document**

☐ No, AI tools have not been used to generate or edit material in this document

Details of use

I carried out the prototype tests and supplied the observation records and peer feedback. ChatGPT helped prepare the written evaluation as listed below. No missing test results were added. Prompt entries are translated summaries, not verbatim transcripts.

Tool	Use	Prompt(s)	Section	Date
ChatGPT (OpenAI)	Helped read and translate the notes I supplied and organise the results. The records came from my prototype tests and peer feedback.	Help develop part of Evaluation 2 using my test results and classmates' feedback.	Results; Appendix A	1 Oct 2026
ChatGPT (OpenAI)	Helped edit the report. These were based on the records I supplied.	Check if there is some grammar problems	Report sections 1-6	3 Oct 2026
ChatGPT (OpenAI)	Helped make the English clearer and more concise, and matched my chosen font and layout. The original test-record images were kept unchanged.	Match my reference document's format, layout and font	report and appendices	5 Oct 2026
ChatGPT (OpenAI)	Helped fill in the AI statement and combine it with the report and appendices in one submission PDF.	Complete the AI declaration using my supplied template .	AI coversheet;	6 Oct 2026

Acknowledgement

I am responsible for the accuracy of the test results, analysis, design decisions and final report.

SPATIAL STORIES

Design Evaluation for Interactive Prototype 2

NAME	STUDENT ID	COURSE
Zixu Yang	48939559	DECO7230

Spatial Stories needs clearer controls, more stable editing and more reliable selection. The three feedback sheets contain generally positive ratings. However, users struggled with resizing, and the task records show that help was needed. The results do not show that users can complete the full task on their own.

01 OBJECTIVE AND VALIDATION METRICS

I tested whether users could create a spatial Story with Quest controllers. The eight actions were to select a photo, add text, add a sticker, move, rotate and resize content, delete an item, then preview and confirm publication. I wanted to find out whether users could follow the controls and on-screen cues without help.

The form used five task codes: independent success (I), success with help (A), failure after an attempt (F), not attempted (NT) and not recorded (NR). Independent completion meant finishing all eight actions without hints during the task. The form also set a 240-second limit. Users rated ease, naturalness and physical comfort from 1 to 5. Return and Undo were optional checks. No overall pass score was set for these ratings.

02 RESULTS

Record	Ease / 5	Naturalness / 5	Comfort / 5	Recorded outcomes
P01	5	4	5	Task codes blank; resizing instructions unclear.
P02	5	4	5	Resize F; six clear A codes; rotation mark unclear.
P03	4	4	3	Sticker and resize F; six other codes A; overall Partial.

Table 1. Ratings and recorded task outcomes from three feedback sheets and two task grids with entries. Appendix A explains how the notes were read. Appendix B contains the original sheets.

All three feedback sheets mention problems with resizing. P01 said the feature and instructions were unclear. P02 found rotation and resizing confusing and could not find the resize control. P03 said resizing was hard to understand, objects rotated while being resized, and objects tilted when grabbed. P03 also failed to select a sticker despite two recorded hints and needed three hints at review/publication. The text task is marked A, but its note says Send could not be selected.

P01 and P02 used preset text. P03's input mode was not recorded. Median ratings were 5 for ease, 4 for naturalness and 5 for comfort. Comfort scores ranged from 3 to 5. All timing and optional-check fields were blank. P04-P06 had no responses and were left out of the analysis. The missing records mean that a completion rate and average task time cannot be calculated.

03 ANALYSIS / INSIGHTS

Resizing controls were hard to understand. All three feedback sheets mention resizing, and both task grids with entries mark it F. P02 could not find a resize button. This suggests that users needed clearer guidance on how to resize content. However, the records do not show whether the problem was unclear instructions, a technical fault, or both. The next test needs to check these possible causes separately.

Objects may need a more stable editing angle. P03 reported unwanted rotation while resizing and tilt while grabbing. This suggests that free movement in 3D may make it harder to arrange a flat Story. Keeping content on one plane could help users control it. This idea comes from one person's feedback, so I need to test it before deciding whether to keep it.

Selection problems blocked later steps. P03 could not select a sticker, had trouble with Send and needed three hints at review/publication. Users need these basic actions to work before they can focus on creating a Story. I should check whether targets are easy to see and select, and whether the system clearly responds. The notes do not identify the technical cause, so I need to check and retest these controls.

High ratings did not mean users could work without help. P02 gave ease a score of 5 despite needing help and failing the resize task. P03 gave ease a score of 4 despite two failed actions. Users may have rated the overall experience after receiving help. The broad questions did not show what they based their scores on. I will therefore read the ratings alongside the task records, rather than use them as proof of success.

04 EVALUATION OF AIMS

Aim	Judgement	Evidence and limits
Creating a Story without help	Not validated	P02 and P03 needed help and failed some actions. P01 has no task outcomes.
Clear manipulation controls	Not met for resizing	All three mention resizing problems. Both recorded resize attempts are F.
Natural and comfortable interaction	Partially supported	Naturalness is 4/5 for all three. Comfort is 5, 5 and 3. P03 reports unwanted tilt and rotation.
Completion within 240 seconds	Uncertain	No start, end or total times were recorded.
Return and Undo	Uncertain	Both optional-check fields are blank, so their success is unknown.

The results support further work on spatial Story creation. They do not show that first-time users can complete the task alone, or that this version is better than IP1. The sample is small. P02 had tried Spatial Stories before, P03 had not, and P01's previous use was not recorded. These differences make the results harder to compare.

05 CONCEPT ITERATION

Spatial Stories will remain a VR editor controlled with Quest controllers. It will keep the floating photo gallery, text and sticker tools, deletion area and publication preview. The next version will add step-by-step guidance for new users and an editing mode that keeps content on a stable plane. Users will select content in space, arrange it at a steady angle, and see which object and action are active. These are planned changes; they have not yet been tested.

Make selection more reliable. P03's problems with the sticker and Send show why this comes first. I will check whether each target is visible and responds to the controller. Where needed, I will make the selectable area larger. Clear visual feedback will show when the controller points at, selects or activates a target. Send and Confirm will stay visible and will be tested separately. Users should be able to activate the right target without someone pointing it out.

Teach one action at a time. All three feedback sheets mention resizing, and one note asks for step-by-step instructions. I will add a practice object with one instruction at a time. Controller diagrams will explain selection and gripping. A short animation will show how to resize with two controllers. A hint will appear when an object is selected. I will then remove the guidance and check whether users can repeat the action on another object.

Keep objects steady during editing. To address P03's tilt and rotation problems, moving or resizing an object will keep it on the same plane by default. Users will rotate it through a separate, clearly labelled action and can reset its angle. Free rotation can remain an option. I will compare accidental rotation and placement accuracy with the current version before choosing the final approach.

Make preview and recovery easier to use. P03 needed three hints at publication, so I will make the steps from editing to preview and confirmation clearer. I will keep Return and Undo as planned features and show when they are available and when an action has worked. Their test fields were blank, so I cannot say they worked well. I will test whether users can find and use them without help.

06 REFLECTION AND NEXT STEPS

The task records and written feedback helped me connect user problems to specific design changes. They also showed gaps in my recording: missing task outcomes and times, unclear hint totals and broad questions. The note that questions were "too open" shows why I need to ask about specific actions. With only three peer feedback records and different levels of experience, I cannot assume the results apply to other users. The comfort ratings also cannot show whether longer use would be comfortable.

Next time, I will trial the form first and record consent, the date and the prototype version. I will separate first-time users from returning users. Everyone will receive the same introduction to the controllers before the timed task. I will record every hint at the step where it happens. I will keep observations, user comments and my own ideas separate. Follow-up questions will ask which control users expected, what made an object tilt, and how they knew an item was selected or published.

Before the next test, I will set an initial target: four of five first-time users should complete all eight actions within 240 seconds without hints during the task. I will also record resize failures, accidental rotation, missed selections and ease ratings for specific actions. Return and Undo will be tested separately. This target is for the next test and is not used to judge this round.

APPENDIX A EVIDENCE TRANSCRIPTION

This appendix records the evidence used in the report. Appendix B contains source pages 1-6 for P01-P03. Source pages 7-12 are blank P04-P06 forms, including the reserve form, so they are not included in the analysis. P01-P03 identify the forms; the forms alone do not confirm who attended.

A means success with help. F means the action was attempted but failed. NR means the field was blank. U means the handwriting was unclear, so no outcome was assumed. A blank hint field is not counted as zero. If a task code and a written note differ, both are reported.

Action	P01	P02	P03	Recorded P03 hints
S01 Photo selection	NR	A	A	NR
S02 Text	NR	A	A*	2
S03 Sticker	NR	A	F	2
S04 Move	NR	A	A	NR
S05 Rotate	NR	U	A	NR
S06 Resize	NR	F	F	1
S07 Delete	NR	A	A	NR
S08 Review and publish	NR	A	A	3

* P03 S02 is marked A, but the note says Send could not be selected. The code remains A. The code and note may describe different moments, but the form does not record the order.

Field	P01	P02	P03
Ease / Naturalness / Comfort	5 / 4 / 5	5 / 4 / 5	4 / 4 / 3
VR experience	NR	Some	Some
Stories use	NR	Creator	Creator
Previous Spatial Stories test	NR	Yes	No
Text input mode	Preset	Preset	NR
Overall outcome checkbox	NR	NR	Partial
Independent completion checkbox	NR	NR	NR
Core prompt total	U	2	NR
Start / end / elapsed time	NR	NR	NR
Consent checkbox	NR	Yes	NR
Recording checkbox	NR	No	NR

P01's total number of prompts is unclear. P02 has a total of 2, but six clear A codes and no hints recorded for individual actions. It is unclear how these entries relate. P03's four filled hint cells add up to 8. Other hint cells and the total are blank, so 8 is only the recorded subtotal. The date, prototype version, practice or task exposure, and optional-check results are missing for all three records.

APPENDIX A FEEDBACK AND SOURCE NOTES

The English descriptions below translate or summarise the feedback. Quotation marks are used only for short phrases originally written in English. The forms do not always show whether a note records the user's own words or the facilitator's summary. Translated feedback is not presented as an exact interview quote.

P01 observation sheet, source page 1. A margin note asks for step-by-step instructions. The facilitator's name is filled in and Preset is selected. Task outcomes, times and overall completion are blank. The total number of prompts is unclear.

P01 feedback sheet, source page 2. Ease 5, naturalness 4 and comfort 5 are marked. The notes mention resizing and include "not clarified" and "instruction not clear". These notes suggest a need for clearer resizing instructions. They do not show that resizing was completed or that the user felt physical discomfort.

P02 observation sheet, source page 3. Photo selection, text, sticker, move, delete and review/publication are clearly marked A. Resize is F. The rotation mark is unclear, so it is recorded as U. The total number of prompts is 2, but no hints are listed for individual steps. Preset text is selected.

P02 feedback sheet, source page 4. Ease 5, naturalness 4 and comfort 5 are marked. The typed Chinese response says rotation and resizing were confusing because the user did not understand how they worked and could not find the resize button. It asks for clearer, more detailed button instructions. A handwritten note says the experience was comfortable. The response about discomfort says none.

Feedback on the questions, source page 4. A separate English note starts with "questions too open" and asks for questions to be more closely linked to the subject/object of the test. The writer is not identified. This is treated as feedback on the questions, not as a comment from a named tutor or an extra participant response.

P03 observation sheet, source page 5. Text is marked A with 2 hints, but the note says Send could not be selected. Sticker selection is F with 2 hints and a note that the sticker could not be selected. Resize is F with 1 hint. Review/publication is A with 3 hints. Photo selection, move, rotate and delete are marked A. The overall result is Partial.

P03 feedback sheet, source page 6. Ease 4, naturalness 4 and comfort 3 are marked. The Chinese notes say resizing was hard to understand, grabbing caused tilt, and rotation during resizing moved content out of its plane. These comments suggest that editing stability needs further testing. They do not confirm a technical fault or explain why comfort was rated 3.

Limits of the records

There are three feedback sheets with responses and two task grids with entries. None shows that all eight actions were completed without help. Blank timing, optional-task and consent fields are treated as missing information. A blank consent box does not prove that consent was refused. These records also do not support a claim that IP2 improved on IP1.

The report follows the six sections in the Interactive Prototype Outcome Evaluation 1 and 2 brief. The full evaluation is on report pages 1-3. The appendices contain the records and notes that support it.

SPATIAL STORIES | IP2a | DECO7230

P01 Observation Record

give instructions stops by stop.

观察记录 | Core task 核心任务

Date 日期 _____ Build 版本 _____ Facilitator 主持人 Adrien
 Session ID 日志编号 _____ Photo 照片 _____

Consent 同意参与 ☐ Yes 是 ☐ No 否 Recording 录像 ☐ Yes 是 ☐ No 否

VR experience 经验 ☐ None 无 ☐ Some 少量 ☐ Frequent 经常

Stories use 使用经历 ☐ None 无 ☐ Viewer 观看 ☐ Creator 制作

Tested Spatial Stories before 曾体验此项目 ☐ No 否 ☐ Yes 是 Details _____

Pre-task practice and any task exposure 计时前练习及任务暴露

Start 开始 _____ End 结束 _____ Elapsed 用时 _____ s Limit 上限 240 s

Codes 代码 I 独立成功 A 提示后成功 F 尝试未成功 NT 未尝试 NR 未记录

Action 操作	Code 结果	Hints 提示数	Errors and notes 错误与观察
S01 Photo selection 选择并近看照片			
S02 Text 添加并放置文字			
S03 Sticker 添加并放置贴纸			
S04 Move 移动一个可编辑元素			
S05 Rotate 旋转一个可编辑元素			
S06 Resize 缩放一个可编辑元素			
S07 Delete 删除一个不需要的元素			
S08 Review and publish 预览并确认发布			

Core prompts 核心任务提示总次数 8 Text 输入方式 ☐ Typed 自输 ☒ Preset 预设

Overall 总结结果 ☐ Complete 完成 ☐ Partial 部分 ☐ Failed 未完成

☐ Interrupted 中断 ☐ NR 未记录 Independent full completion 全程独立 ☐ Yes 是 ☐ No 否 ☐ Unknown 未知

Interruption or critical issue 中断或关键问题及时间

SPATIAL STORIES | IP2a | DEC07230

P01 Feedback and Help Record

评分与提示记录 | Pair with this participant's observation page

Date 日期 _____ Build 版本 _____

Post task questions 测试后问题

How easy was the overall task? 整体任务有多容易?

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☒ NR 未回答 ☐

1 Very difficult 很难 · 3 Neither 中等 · 5 Very easy 很容易

How natural did manipulating content in this VR space feel? 在这个 VR 空间操作内容感觉有多自然?

1 ☐ 2 ☐ 3 ☐ 4 ☒ 5 ☐ NR 未回答 ☐

1 Very unnatural 很不自然 · 3 Neither 中等 · 5 Very natural 很自然

How physically comfortable was the experience? 这次体验在身体上有多舒适?

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☒ NR 未回答 ☐

1 Very uncomfortable 很不适 · 3 Neither 中等 · 5 Very comfortable 很舒适

Most confusing action 最困惑的操作及原因

One change you would make 最想修改的一点

Discomfort or reach/readability issue 不适或够不到/看不清的问题

not clarified.

resizing feature is

instruction not clear

Prompt and event notes 提示与事件记录

Time + stage + exact cue or observed event. Mark quotes Q and paraphrases P.

写时间、步骤、具体提示或观察。原话标 Q，转述标 P；不要把推测写成原话。

Optional checks 额外检查

Run after core questions in a fresh editing opportunity; leave core totals unchanged.

在核心提问后另行进入编辑状态测试，不改核心任务计数。

O1 Return and revisit 返回并重选原照片 Code _____ Prompts _____

O2 Recover latest deletion 撤销最近删除 Code _____ Prompts _____

O2: immediately after a new deletion ask "Can you get that item back?" Default window about 5 s. NT if not run.

另删一个物体后立即问“你能把它找回来吗？”默认约 5 秒有效，未测记 NT。

Evidence file(s) 证据文件 _____ Reset 重置 ☐

P02 Observation Record

观察记录 | Core task 核心任务

Date 日期 _____ Build 版本 _____ Facilitator 主持人 Ad

Session ID 日志编号 _____ Photo 照片 _____

Consent 同意参与 ☒ Yes 是 ☐ No 否 Recording 录像 ☐ Yes 是 ☒ No 否

VR experience 经验 ☐ None 无 ☒ Some 少量 ☐ Frequent 经常

Stories use 使用经历 ☐ None 无 ☐ Viewer 观看 ☒ Creator 制作

Tested Spatial Stories before 曾体验此项目 ☐ No 否 ☒ Yes 是 Details _____

Pre-task practice and any task exposure 计时前练习及任务暴露 _____

Start 开始 _____ End 结束 _____ Elapsed 用时 _____ s Limit 上限 240 s

Codes 代码 I 独立成功 A 提示后成功 F 尝试未成功 NT 未尝试 NR 未记录

Action 操作	Code 结果	Hints 提示数	Errors and notes 错误与观察
S01 Photo selection 选择并近看照片	A		
S02 Text 添加并放置文字	A		
S03 Sticker 添加并放置贴纸	A		
S04 Move 移动一个可编辑元素	A		
S05 Rotate 旋转一个可编辑元素	A		
S06 Resize 缩放一个可编辑元素	F		
S07 Delete 删除一个不需要的元素	A		
S08 Review and publish 预览并确认发布	A		

Core prompts 核心任务提示总次数 2 Text 输入方式 ☐ Typed 自输 ☒ Preset 预设

Overall 总结结果 ☐ Complete 完成 ☐ Partial 部分 ☐ Failed 未完成

☐ Interrupted 中断 ☐ NR 未记录 Independent full completion 全程独立 ☐ Yes 是 ☐ No 否 ☐ Unknown 未知

Interruption or critical issue 中断或关键问题及时间 _____

SPATIAL STORIES | IP2a | DECO7230

P02 Feedback and Help Record

评分与提示记录 | Pair with this participant's observation page

Date 日期 _____ Build 版本 _____

Post task questions 测试后问题

How easy was the overall task? 整体任务有多容易?

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☒ NR 未回答 ☐

1 Very difficult 很难 · 3 Neither 中等 · 5 Very easy 很容易

How natural did manipulating content in this VR space feel? 在这个 VR 空间操作内容感觉有多自然?

1 ☐ 2 ☐ 3 ☐ 4 ☒ 5 ☐ NR 未回答 ☐

1 Very unnatural 很不自然 · 3 Neither 中等 · 5 Very natural 很自然

How physically comfortable was the experience? 这次体验在身体上有多舒适?

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☒ NR 未回答 ☐

1 Very uncomfortable 很不适 · 3 Neither 中等 · 5 Very comfortable 很舒适

好舒服

Most confusing action 最困惑的操作及原因 旋转和缩放 因为我不懂原理, 缩放没有找到按键

One change you would make 最想修改的一点 按键指示不够详细易理解

Discomfort or reach/readability issue 不适或够不到/看不清的问题 没有

Prompt and event notes 提示与事件记录

Time + stage + exact cue or observed event. Mark quotes Q and paraphrases P.

写时间、步骤、具体提示或观察。原话标 Q, 转述标 P; 不要把推测写成原话。



questions too open

more about subject

more linked to the

object.

Optional checks 额外检查

Run after core questions in a fresh editing opportunity; leave core totals unchanged.

在核心提问后另行进入编辑状态测试, 不改核心任务计数。

O1 Return and revisit 返回并重选原照片 Code _____ Prompts _____

O2 Recover latest deletion 撤销最近删除 Code _____ Prompts _____

O2: immediately after a new deletion ask "Can you get that item back?" Default window about 5 s. NT if not run.

另删一个物体后立即问“你能把它找回来吗?”默认约5秒有效,未测记NT。

Evidence file(s) 证据文件 _____ Reset 重置 ☐

P03 Observation Record

观察记录 | Core task 核心任务

Date 日期 _____ Build 版本 _____ Facilitator 主持人 Ad

Session ID 日志编号 _____ Photo 照片 _____

Consent 同意参与 ☐ Yes 是 ☐ No 否 Recording 录像 ☐ Yes 是 ☐ No 否

VR experience 经验 ☐ None 无 ☒ Some 少量 ☐ Frequent 经常

Stories use 使用经历 ☐ None 无 ☐ Viewer 观看 ☒ Creator 制作

Tested Spatial Stories before 曾体验此项目 ☒ No 否 ☐ Yes 是 Details _____

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Action 操作	Code 结果	Hints 提示数	Errors and notes 错误与观察
S01 Photo selection 选择并近看照片	A		
S02 Text 添加并放置文字	A	2	无法选中发送
S03 Sticker 添加并放置贴纸	F	2	完全无法选中
S04 Move 移动一个可编辑元素	A		
S05 Rotate 旋转一个可编辑元素	A		
S06 Resize 缩放一个可编辑元素	F	1	
S07 Delete 删除一个不需要的元素	A		
S08 Review and publish 预览并确认发布	A	3	

Core prompts 核心任务提示总次数 7 Text 输入方式 ☐ Typed 自输 ☐ Preset 预设

Overall 总结结果 ☐ Complete 完成 ☒ Partial 部分 ☐ Failed 未完成

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Interruption or critical issue 中断或关键问题及时间

SPATIAL STORIES | IP2a | DECO7230

P03 Feedback and Help Record

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1 Very uncomfortable 很不适 · 3 Neither 中等 · 5 Very comfortable 很舒适

Most confusing action 最困惑的操作及原因

One change you would make 最想修改的一点

Discomfort or reach/readability issue 不适或够不到/看不清的问题

缩放会旋转



放大缩小 → 不是很直观。不在平面上

grab 的时候会倾斜 → 影响平面的感觉。

Prompt and event notes 提示与事件记录

Time + stage + exact cue or observed event. Mark quotes Q and paraphrases P.

写时间、步骤、具体提示或观察。原话标 Q，转述标 P；不要把推测写成原话。

Optional checks 额外检查

Run after core questions in a fresh editing opportunity; leave core totals unchanged.

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Evidence file(s) 证据文件 _____ Reset 重置 ☐